

Question	Answer	Marks
5(a)	<u>sum of</u> current(s) into junction = <u>sum of</u> current(s) out junction or (algebraic) sum of current(s) at a junction is zero	B1
5(b)(i)	line with negative gradient	C1
	line from 0 °C to 100 °C, starting at (0, R_0) with negative gradient throughout and never reaching $R = 0$	A1
5(b)(ii)	(resistance of T decreases so) total resistance (of circuit) decreases	B1
	current in cell increases (so p.d. across internal resistance increases)	B1
	terminal p.d. decreases (and resistance of R is constant) so current in R decreases	B1
5(c)(i)	p.d. across $r = 1.50 - (6.00 \times 0.200)$ (= 0.30 V)	C1
	current in cell = $0.30 / 0.12$ = 2.5 A	A1
5(c)(ii)	current in thermistor = $2.5 - 0.200$ (= 2.3 A)	C1
	resistance of T = $(6.00 \times 0.200) / 2.3$ = 0.52 Ω	A1

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